METRIC

A-A-50623
16 September 1994
SUPERSEDING
MIL-A-17196E (IN PART)
7 October 1977
(See Note 1)

COMMERCIAL ITEM DESCRIPTION

ALARM SWITCHBOARD TYPE IC/SM

The General Services Administration has authorized the use of this commercial item description as a replacement for Type IC/SM Switchboards in MIL-A-17196E(SH).

Abstract. This Commercial Item Description covers requirements for IC/SM Alarm Switchboards which will enable one operator to monitor equipment functions or environmental conditions of remote equipment. IC/SM Alarm Switchboards shall be capable of being connected to mechanical, electrical, electronic, electromechanical and environmental-sensing devices. Alarm switchboards shall be such that remote sensors located at each monitor point will provide switch closure to signal the alarm detection circuit of a fault condition. In addition, this circuit will provide a supervisory current loop which provides for operational confidence in the remote sensor circuit. Alarm switchboards shall include provisions for remote audible and visual indication of alarm conditions in addition to the central displays.

Salient characteristics.

1. **Integral Components**. Type IC/SM Alarm Switchboards shall be composed of an IC/SM alarm panel, such as shown in Figures 1 thru 8, and IC/M alarm modules mounted within the alarm panel.

2. **General requirements**

a. **Power requirements**. The alarm switchboards shall be designed for operation from a normal source of power conforming to Type I, 115 V, 60 Hz, 1 phase. A rectifier shall be provided to supply power for the line circuits, lamp flasher, ground detector lamps and pilot lamp. The supervisory voltage shall be 5 +1/-0 Vdc, full wave, rectified (unfiltered).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent by letter to: Commander, Naval Sea System Command, Attn: 03R42, 2531 Jefferson Davis Highway, Arlington, VA 22242-5160.

AMSC N/A

FSC: 6320

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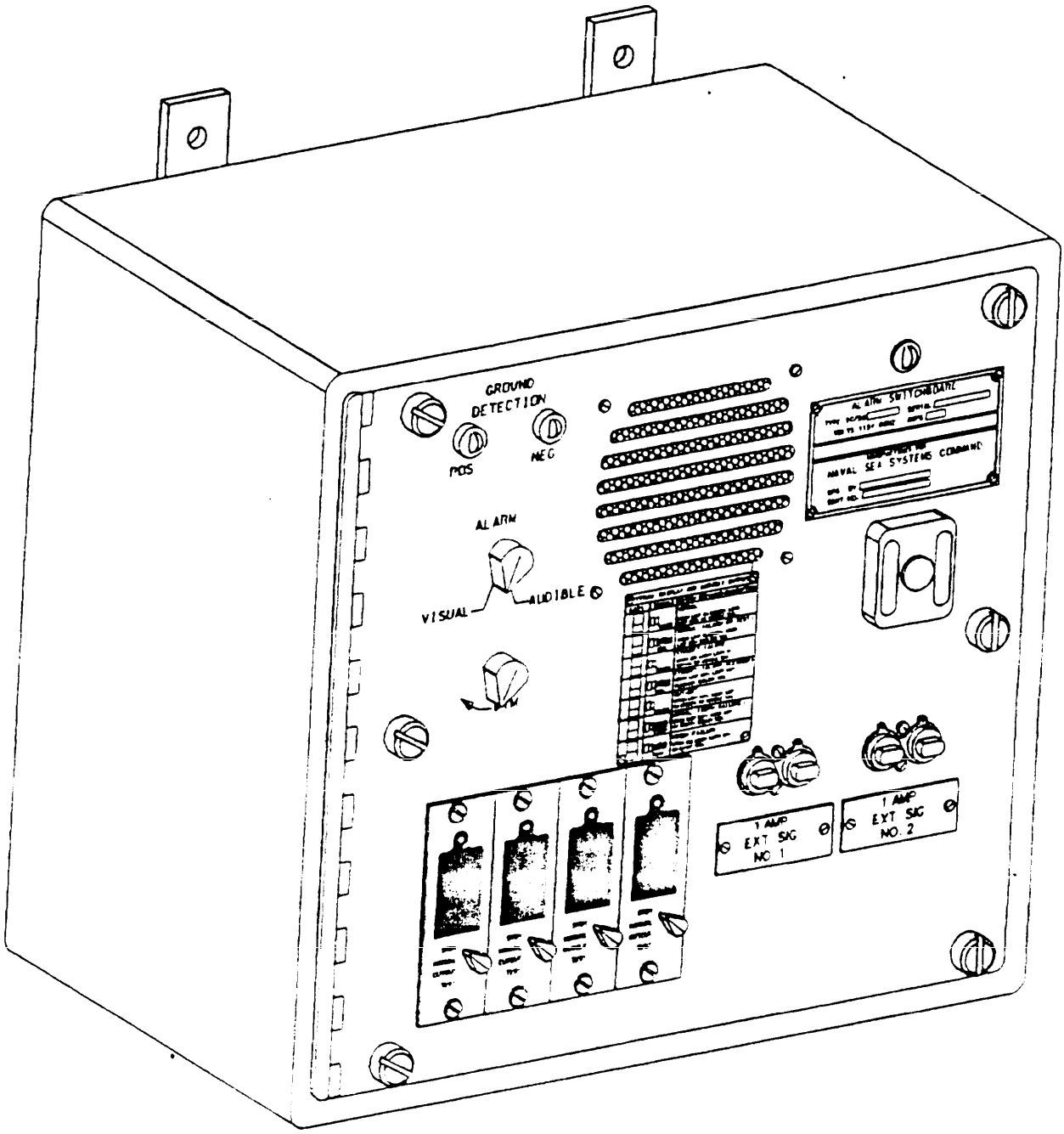


FIGURE 1 - IC/SM-4
4 LINE SWITCHBOARD

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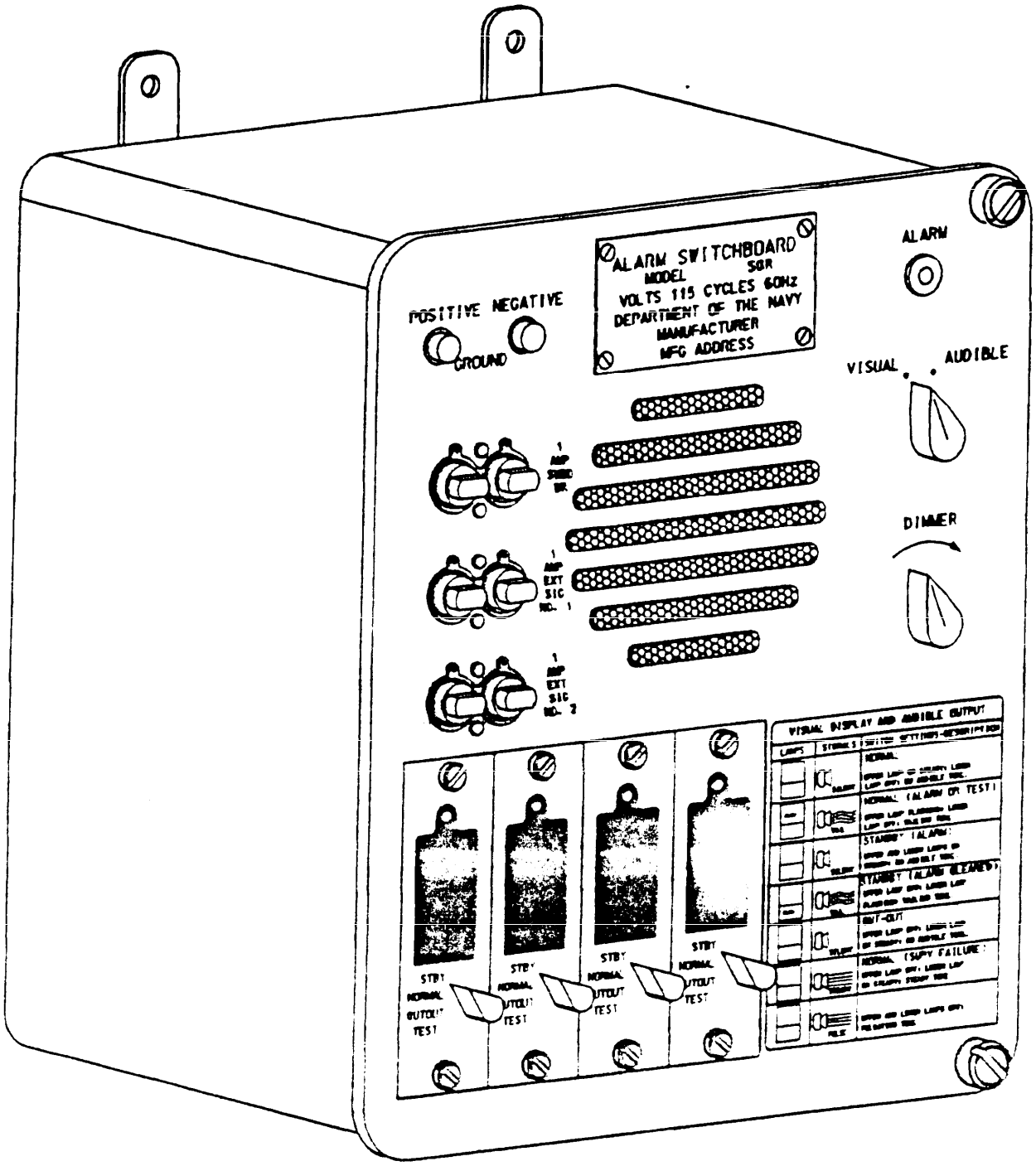


FIGURE 2 - **IC/SM-4**
4 LINE SWITCHBOARD

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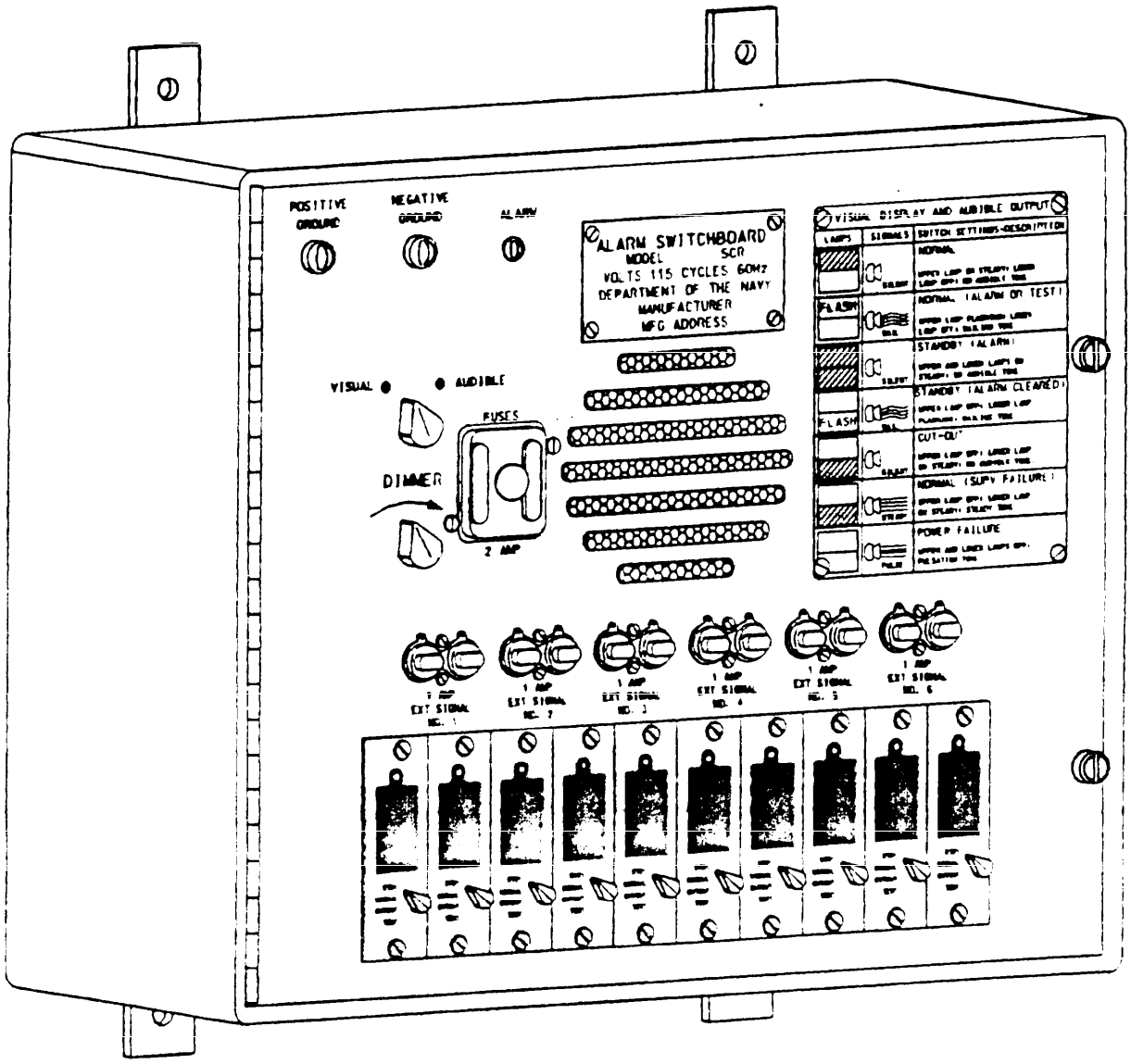


FIGURE 3 - **IC/SM-10**
10 LINE SWITCHBOARD

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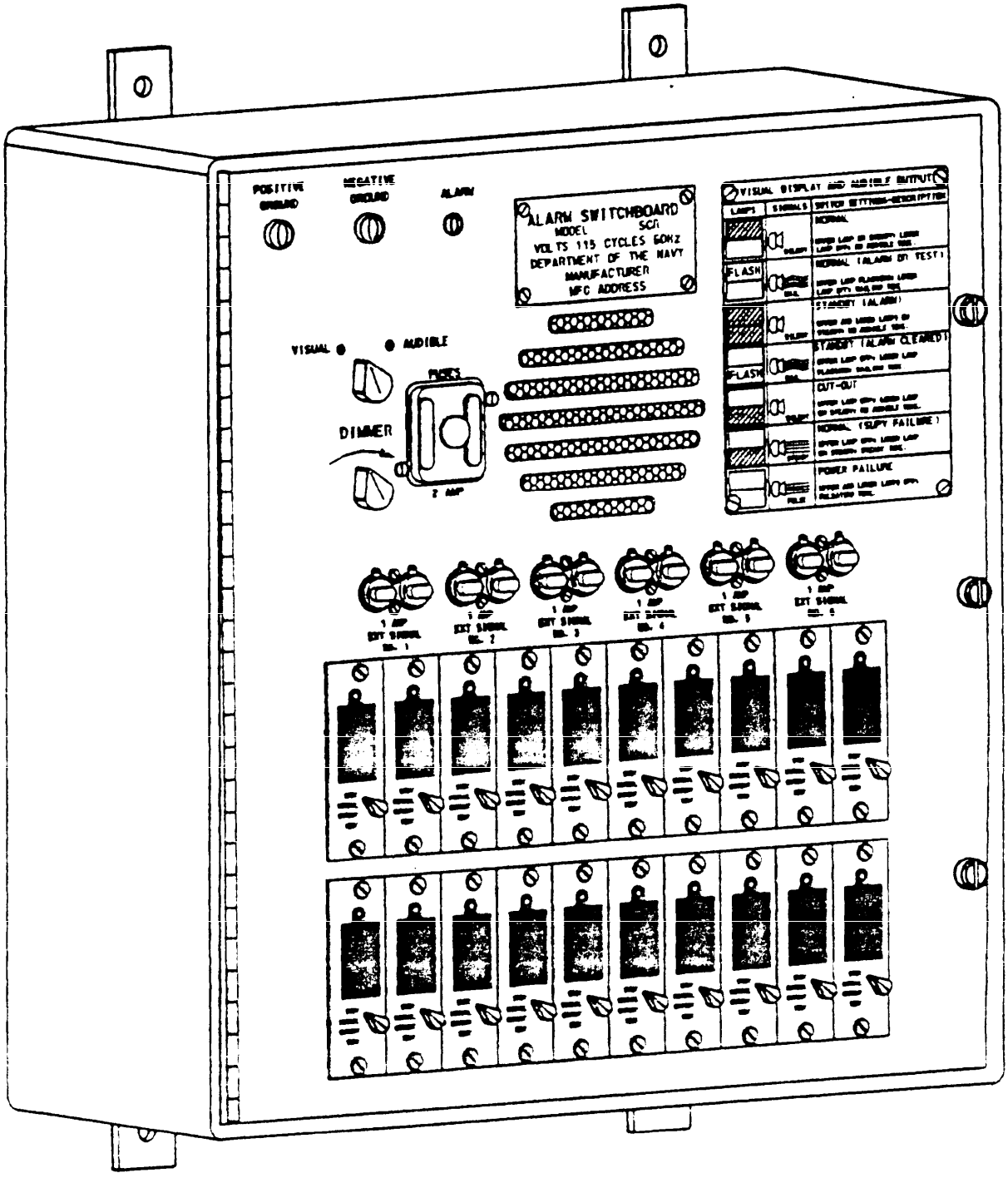


FIGURE 4 - IC/SM-20
20 LINE SWITCHBOARD

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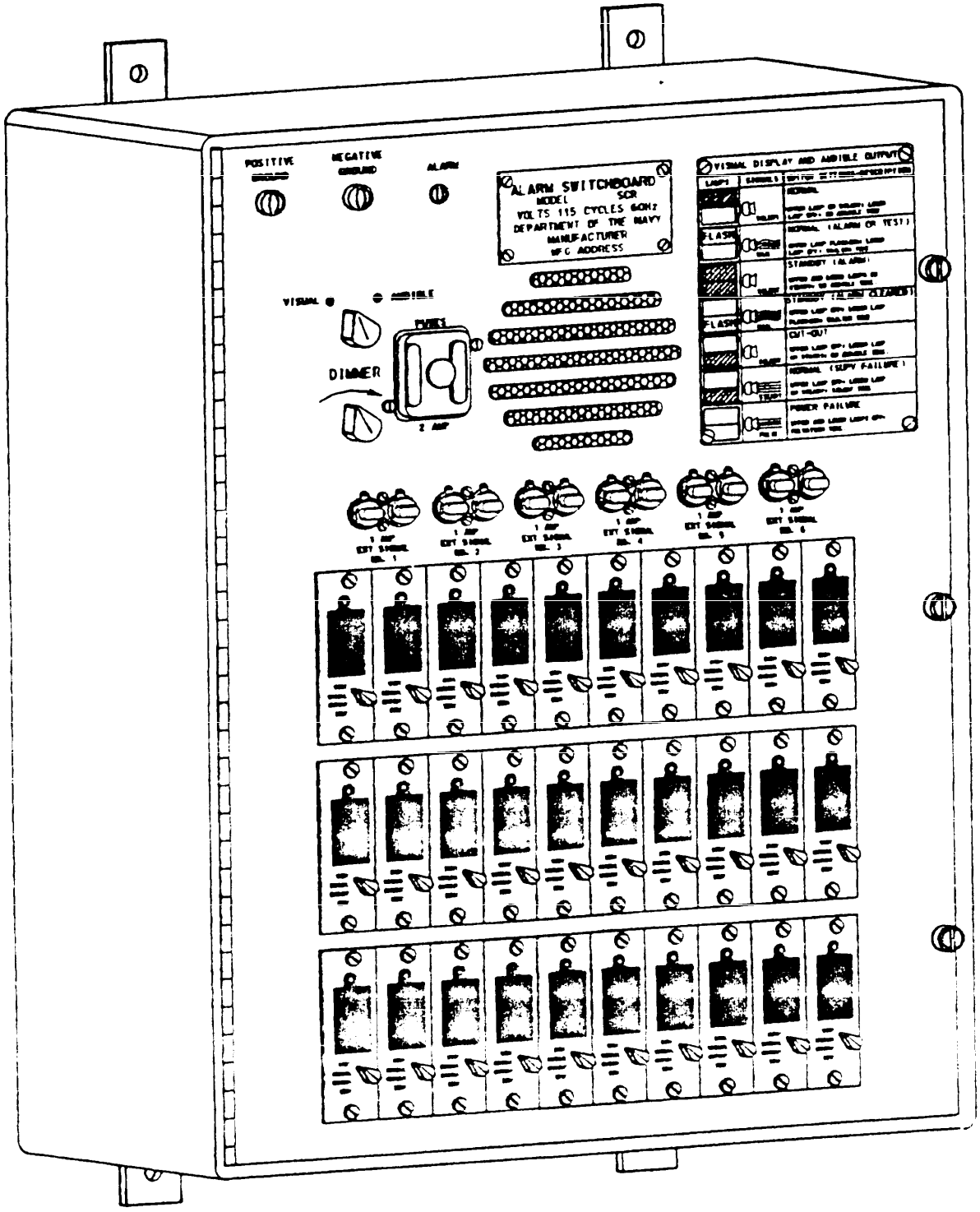


FIGURE 5 - IC/SM-30
30 LINE SWITCHBOARD

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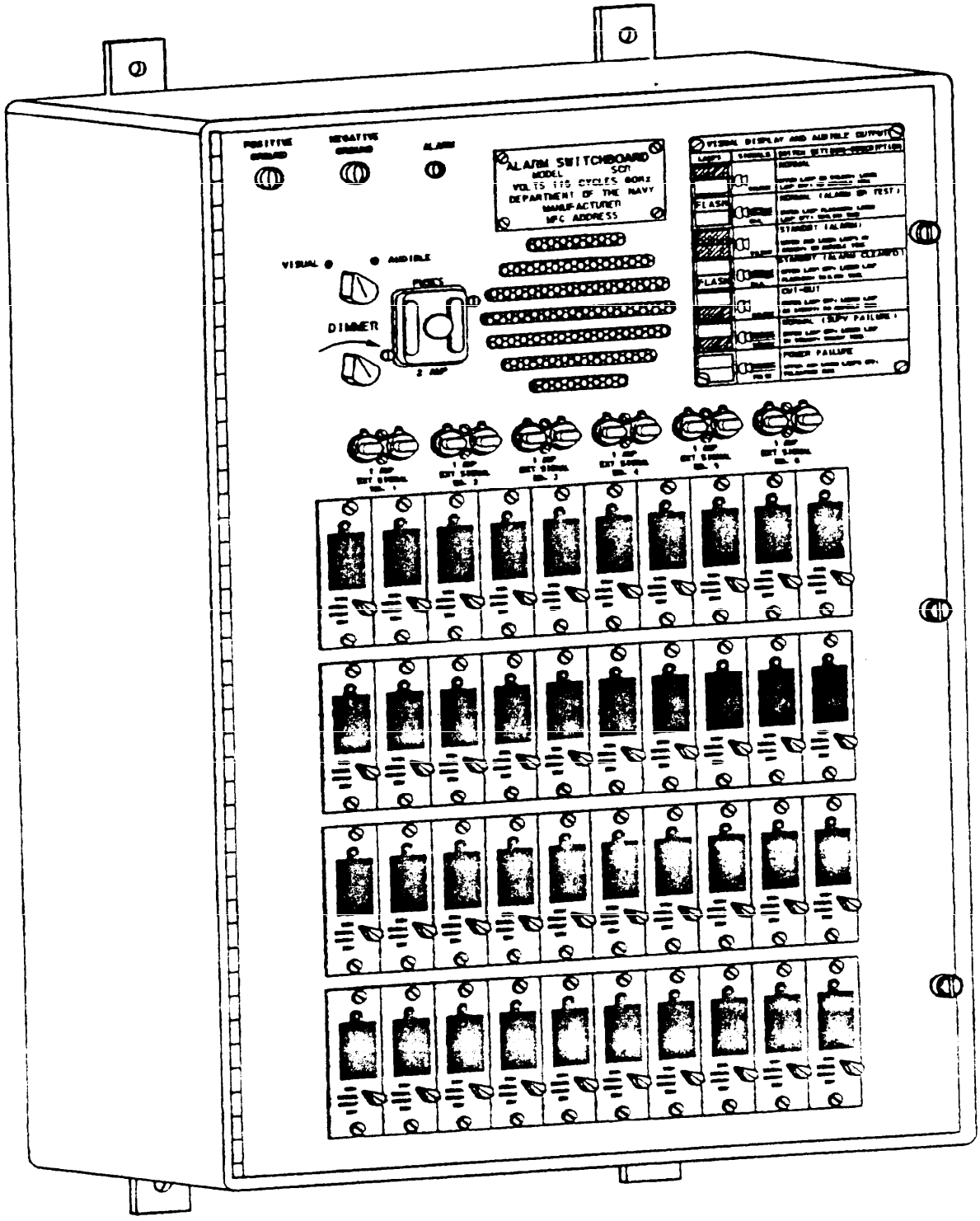


FIGURE 6 - IC/SM-40
40 LINE SWITCHBOARD

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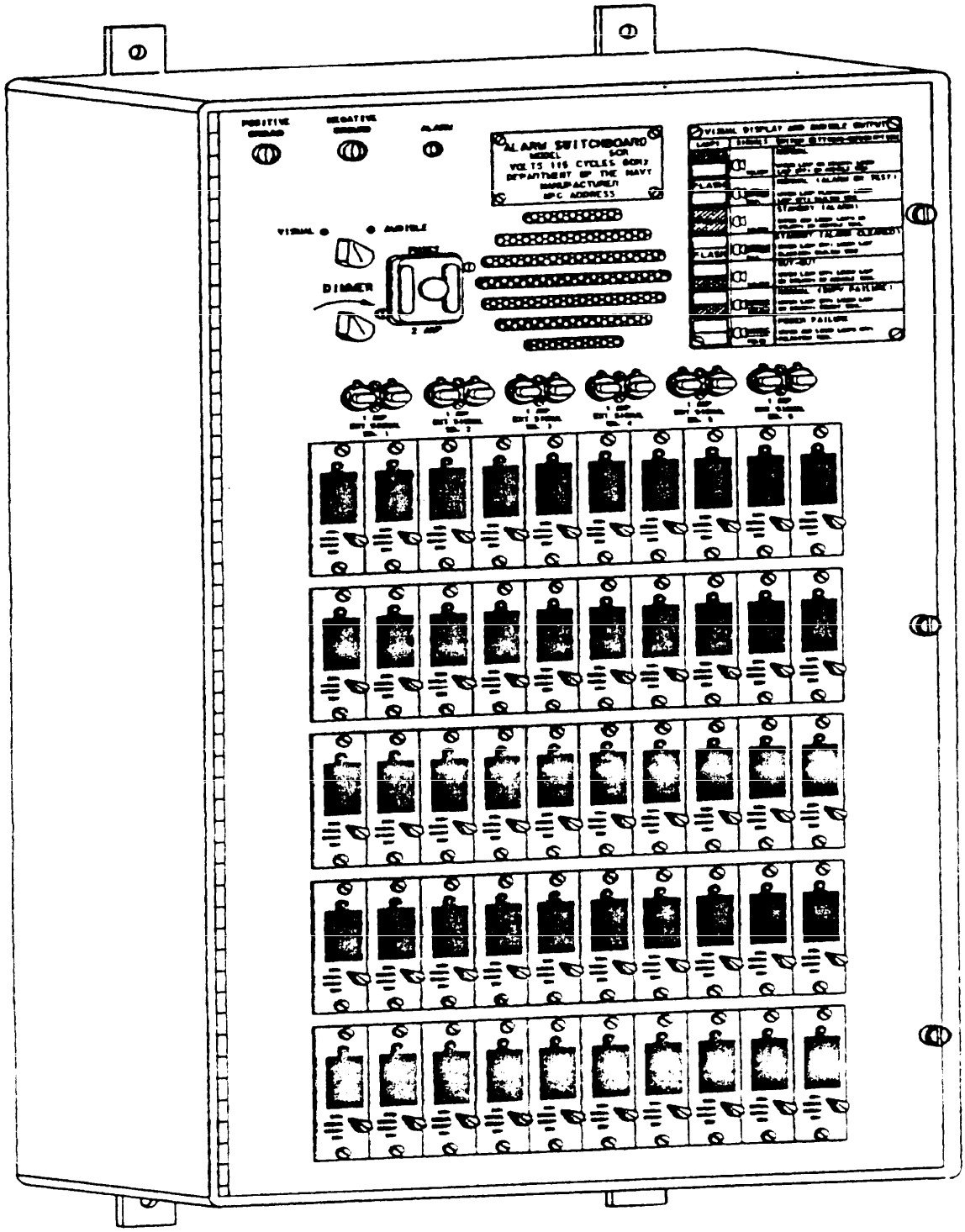


FIGURE 7 - IC/SM-50
50 LINE SWITCHBOARD

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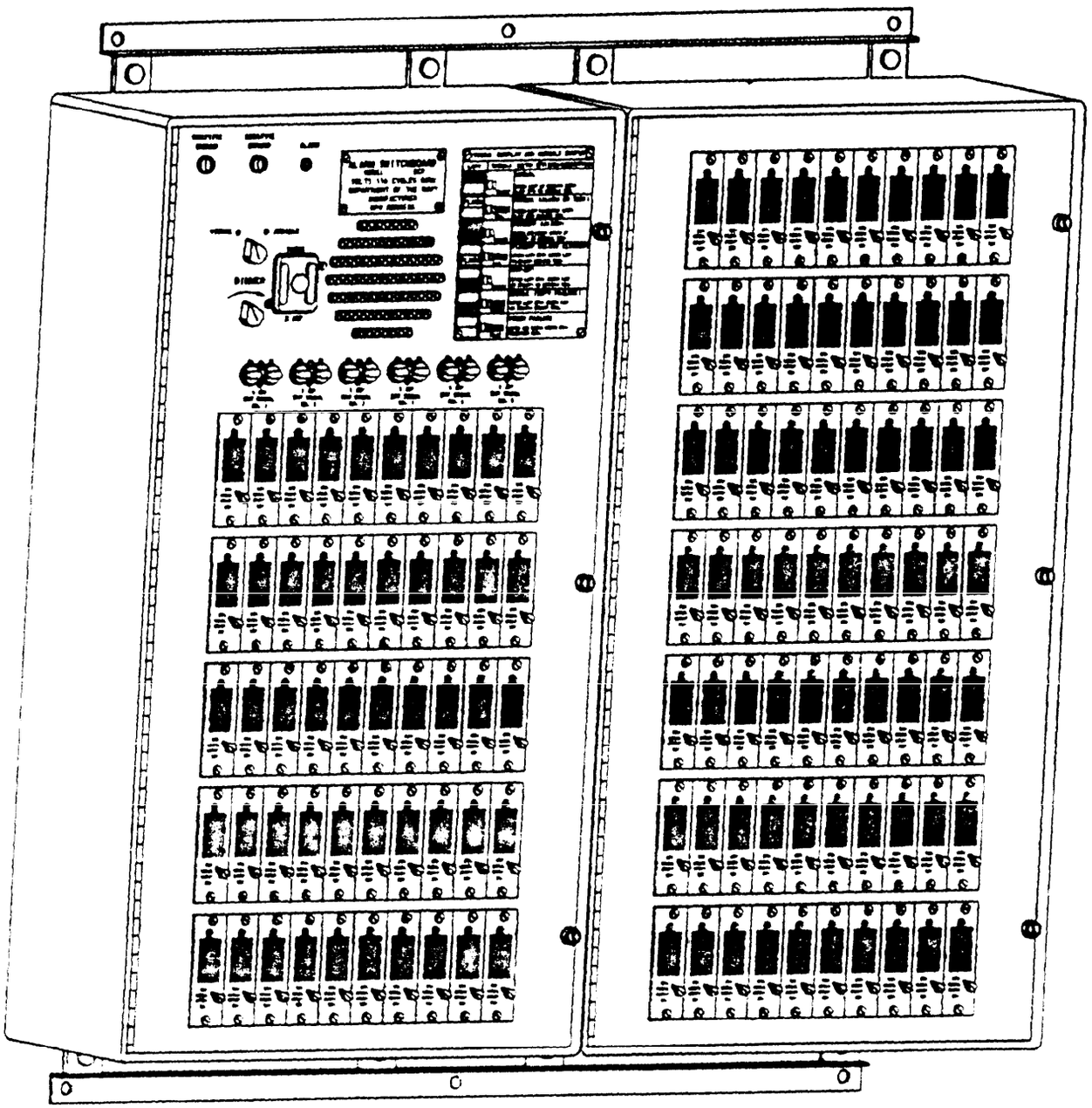


FIGURE 8 - IC/SM-120
120 LINE SWITCHBOARD

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b. **Prohibited materials.** The materials listed below shall not be used in the construction of IC/SM Alarm Switchboards.

(1) Flammable materials
(2) Asbestos; asbestos compounds: and asbestos-filled molding compounds
(3) Lithium and lithium compounds
(4) Magnesium or magnesium alloy
(5) Zinc or Zinc alloy
(6) Carcinogens
(7) Radioactive material
(8) Polychlorinated Biphenyl (PCB)
(9) Polyvinyl Chloride (PVC), except where used for component leads
(10) Mercury or its compounds and amalgams
(11) Cadmium

c. **Preservation.** All parts shall be of suitable corrosion resistance material or materials treated in a satisfactory manner to render them adequately resistant to corrosion caused by sea water.

d. **Painting.** Sheet metal and cast enclosures shall be completely painted, in accordance with the ordering data, in accordance with best commercial practices utilizing latex or epoxy based paints. Composite enclosures (if used) normally shall not be painted.

e. **Temperature and humidity.** Alarm switchboards and alarm modules shall operate without mechanical or electrical damage in any ambient temperature between the limits of 0 and 50° C (Centigrade) and with a relative humidity of 95 percent.

f. **Shock and vibration.** For naval procurement all switchboard units shall satisfactorily withstand, without mechanical or electrical damage, Grade (B), Type A, Class I, Light Weight shock in accordance with MIL-S-901, and vibration testing in accordance with MIL-STD-167, Type 1, Vital.

g. **Electromagnetic Interference (EMI).** For naval procurement switchboard units shall satisfactorily meet the requirements of MIL-STD-461 as demonstrated following the procedures of MIL-STD-462.

h. **Reliability.** IC/SM Alarm Switchboards and IC/M Alarm Modules shall have a Mean Time Between Failure (MTBF) of at least 30K Hours.

3. **IC/M Alarm Module.** Each remote sensing device in an alarm circuit shall be monitored by a type IC/M module, constructed in accordance with FIGURE 9. Each module shall include one set of electrically isolated, normally open contacts which shall close on alarm condition. They shall be suitable for making and breaking an external resistive load of at least 3A at 28 Vdc and shall not be in the completed circuit when the module is operated in the test mode. Upon closure of an alarm sensor, momentary or otherwise, each module shall latch-in and shall not clear even if the alarm sensor re-opens. The switch shall have to be moved to the STANDBY position in order to silence the audible alarm. An alarm condition cleared indication shall be given in the STANDBY switch position alerting the operator to return the switch to NORMAL.

a. **Moisture proofing.** All electronic PC board components shall be conformal coated to ensure operability in a moisture laden environment.

b. **Weight and size.** The maximum weight of the IC/M Module shall not exceed 0.500 Kg nor the size shown in FIGURE 9.

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4. **IC/SM Switchboard.**

a. **Alarm section.** The common alarm section shall have the following parts mounted on or in the face of the panel:

Type IC/SM
(1) Audible speaker.
(2) Module lamp dimmer control.
(3) Ground lights (two).
(4) Switchboard identification plate.
(5) Silent alarm switch.
(6) Silent alarm lamp.
(7) Main power fuse indicator.
(8) Extension alarm blown fuse indicator.

b. **Ground Indicators.** The ground indicators shall sense an absolute ground between the terminal board and the enclosure, with one light connected to indicate a positive ground, and the other connected to indicate a negative ground.

c. **Supervisory current.** A continuous supervisory alarm current, not to exceed 1.2 milliamperes (ma), shall flow through each alarm line module. The circuit for the determination of the normal supervisory current shall include a current limiting resistor, installed in the most electrically remote sensing device in the circuit. A single line alarm module shall trigger a supervisory alarm for supervisory current at 0.1 ma, +0.2 ma/-0.05 ma, but no supervisory alarm shall be generated above 0.3 ma. The module shall trigger a normal alarm for 4.0 ma and higher current flow.

d. **Enclosure.** All switchboards shall be enclosed in a dripproof metal, or composite, cabinet. Each panel shall be arranged to hinge out from the left side, such that all parts shall be accessible from the front of the cabinet. Terminal strips for connecting the ship cables shall be provided in the rear of the enclosure. All flexing cables to hinged panels shall be designed with a maximum amount of slack to distribute bending over the longest practical length of cable. The type of conductors, and the method of cabling the conductors shall be specifically designed for flexing service. Electrical connectors for the sensor lines shall be capable of accepting a bare wire connection from the sensor, without requiring the use of plugs or pins.

e. **Weight and size.** The maximum weight and size of IC/SM Alarm Switchboards shall be as indicated in TABLE I.

5. **SWITCHBOARD LABEL PLATE.** A permanently installed information plate, as illustrated in FIGURE 10, shall be provided on the front of each switchboard in a readily apparent location to identify for an operator the following visual displays and audible outputs of the switchboard. Lights, when required to flash, shall flash at approximately 80 times per minute.

a. **Normal (No-alarm condition).** Upper light on steady, lower light out. (Switch in normal position).

b. **Alarm condition.** Upper light flashing, lower light out with circuitry completed for wailing audible tone. (Switch in normal position)

c. **Standby (alarm condition existing).** Switch turned to standby (30 degrees right from NORMAL position). Upper and lower lights on steady, audible tone circuit open.

d. **Standby (Switch in STANDBY position with alarm condition cleared).** Upper light out, lower light flashing, circuit completed for wailing audible tone. (Operator shall return switch to normal position to silence the audible tone).

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TABLE I
IC/SM ALARM SWITCHBOARD CONFIGURATIONS

Model	WING CORP. DRAWING NUMBER	DYNALEC CORP. DRAWING NUMBER	MAXIMUM DIMENSIONS			
			WIDTH (Cm)	HEIGHT (Cm)	DEPTH (Cm)	WEIGHT (Kg)
IC/SM-4	3448-120	62404-200	38.1	35.6	23.8	12.0
IC/SM-10	3455-210	62404-304	45.7	38.1	24.4	17.0
IC/SM-20	3455-220/270	62404-305	45.7	50.2	24.4	24.0
IC/SM-30	3455-230	62404-306	45.7	62.2	24.4	30.0
IC/SM-40	3455-240	62404-307	45.7	74.3	24.4	36.0
IC/SM-50	3455-250	62404-308	45.7	86.4	24.4	43.0
IC/SM-60	3455-270	62404-311	91.4	50.2	27.6	49.5
IC/SM-80	3455-230/280	62404-301	91.4	62.2	27.6	62.0
IC/SM-100	3455-240/290	62404-302	91.4	74.3	27.6	73.0
IC/SM-120	3455-250/300	62404-303	91.4	86.4	27.6	85.0

e. **Supervisory failure**. Upper light out, lower light on steady, the circuitry completed to steady audible tone. (Switch in normal position). To silence audible alarm, the operator shall switch to cut-out position.

f. **Cut-out**. With the switch in the cut-out position (30 degrees left from NORMAL position, both sides of the line shall be opened and the upper light shall be out and the lower light shall be on steady. Audible tone circuit shall be open.

g. **Test condition**. With the switch in the test (60 degrees left from NORMAL) position, the line resistor shall be shorted out for testing operation of the switchboard. The indications shall be the same as 5.b.

h. **Power failure**. Upper and lower lights shall be out. (The module shall have no function to establish connections for an audible alarm, but the switchboard shall make connections for a pulsating, yelping sound).

6. **Mounting**. Type IC/SM, switchboard cabinets shall be arranged for bulkhead mounting equivalent to that shown in Figure 11. Mounting dimensions shown are maximum allowable.

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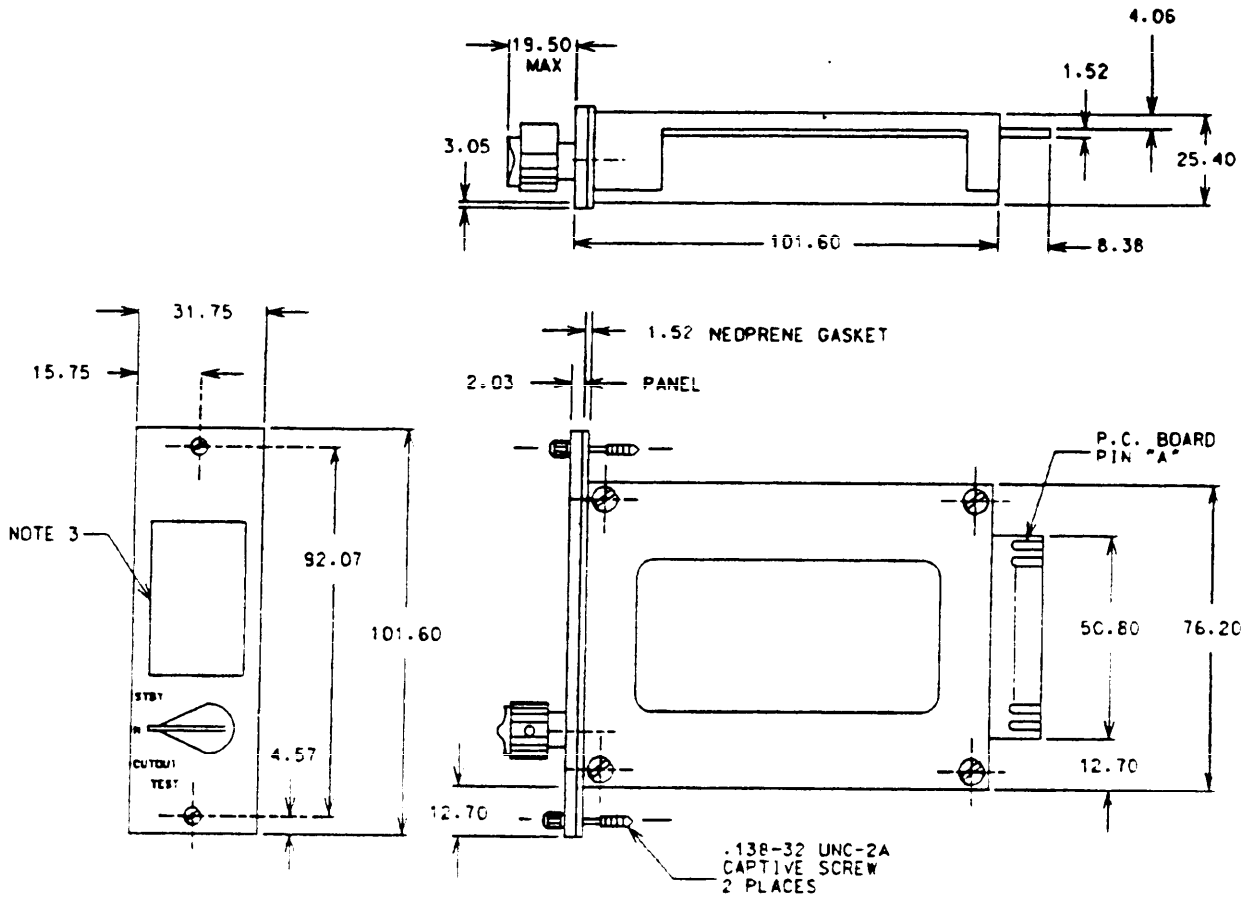


FIGURE 9- **ALARM MODULE. IC/M**

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS, EXCEPT FOR CAPTIVE SCREWS.

2. UNLESS OTHERWISE SPECIFIED HEREIN, TOLERANCES SHALL BE AS FOLLOWS:
   TWO PLACE DECIMALS: ± .02
   THREE PLACE DECIMALS: ± .005

3. SEE CUSTOMERS ORDER FOR LEGEND REQUIREMENTS. INFORMATION TO BE PROCESSED ON PHOTO SENSITIVE MYLAR (ETCHED TYPE).

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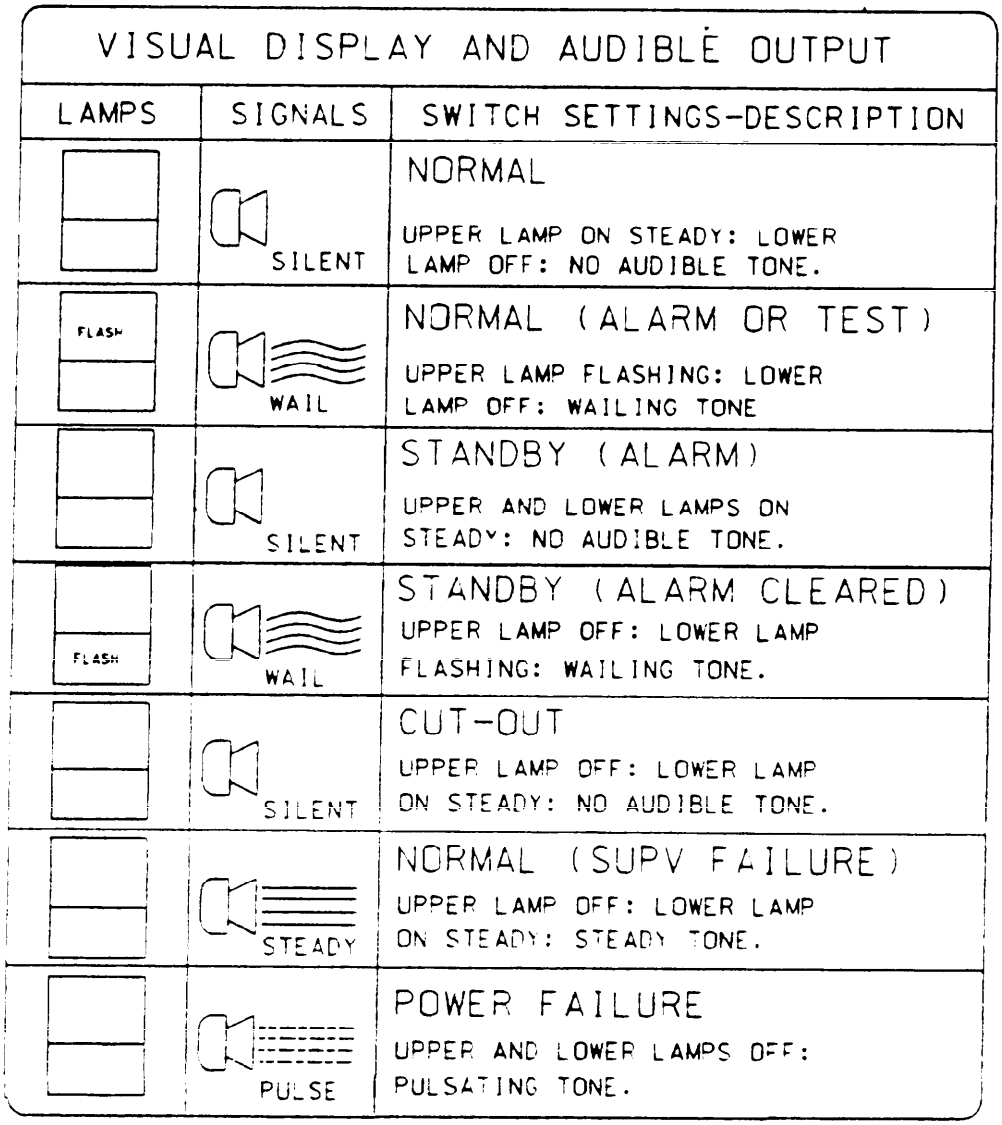
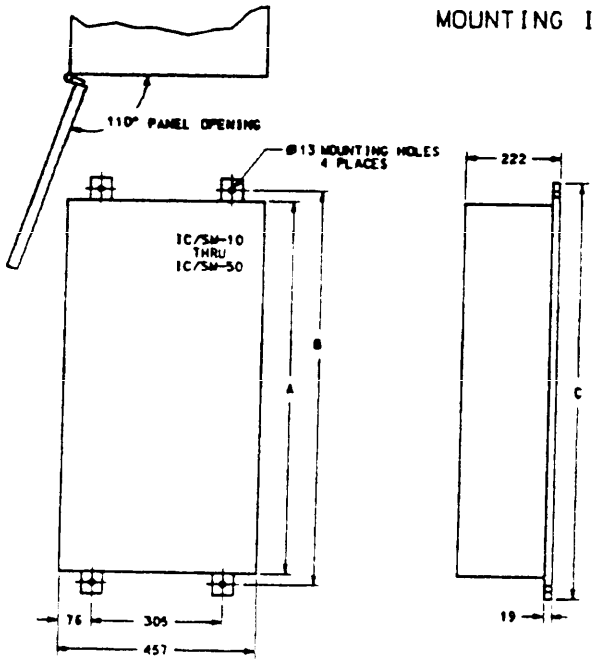


FIGURE 10 - **INFORMATION PLATE**

STEADY AND FLASHING DISPLAYS ON THE MODULE LAMPS, WITH AUDIBLE COMMON ALARM SIGNALS, INDICATE THE STATUS OF EACH REMOTE SENSOR.

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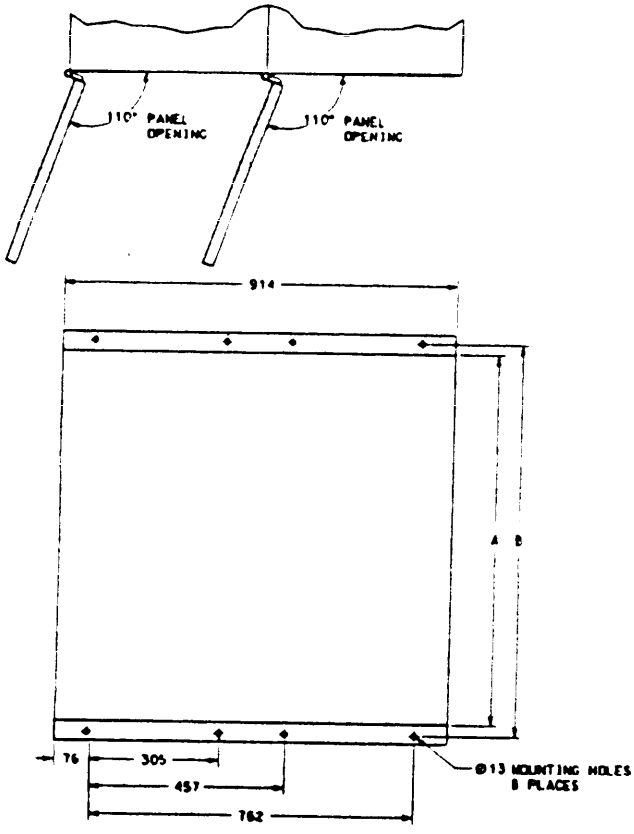
MOUNTING INFORMATION



ALARM SWITCHBOARD IC/SM-10 THROUGH IC/SM-50			
SWITCHBOARD SIZE	MOUNTING DIMENSIONS		
	A	B	C
IC/SM-10	381	432	483
IC/SM-20	502	552	603
IC/SM-30	622	673	724
IC/SM-40	743	794	845
IC/SM-50	864	914	965

KEY

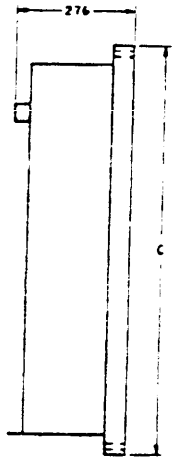
A = HEIGHT OF CABINET
B = DISTANCE BETWEEN RAIL MOUNTING HOLES
C = OVERALL HEIGHT



ALARM SWITCHBOARD IC/SM-60 THROUGH IC/SM-120			
SWITCHBOARD SIZE	MOUNTING DIMENSIONS		
	A	B	C
IC/SM-60	502	552	591
IC/SM-80	622	673	711
IC/SM-100	743	794	832
IC/SM-120	864	914	953

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS



IC/SM-60 THROUGH IC/SM-120 ARE FACTORY-MOUNTED SIDE-BY-SIDE ON A FRAME FOR BULKHEAD MOUNTING.

FIGURE 11 - IC/SM SWITCHBOARD

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reprocessed to become a source of raw materials, as distinguished from virgin

Quality assurance provisions.

1. **Contractor certification.** The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices in accordance with ANSI/ASQC Q-92 (ISO 9002), and ANSI/ASQC Q-93 (ISO 9003). The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

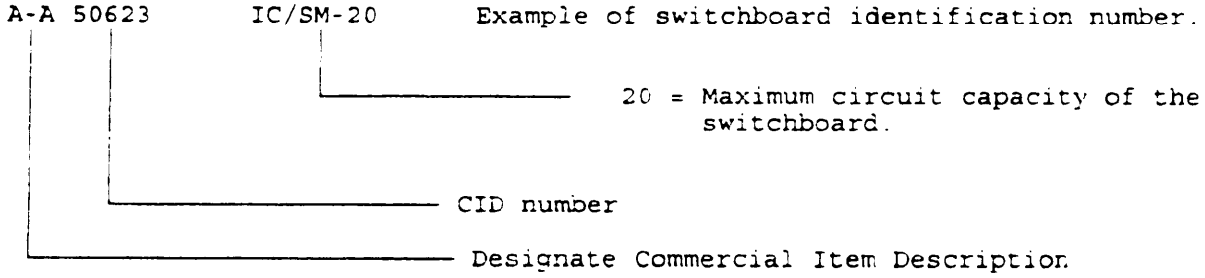
2. **Regulatory requirements.** The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable. Switchboard equipment shall not expose personnel to toxic substances in excess of the threshold limit values in accordance with ACGIH ISBN O-936-712-39-2.

3. **Preservation, packaging, packing, labeling, and marking.** The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Notes.

1. **Supersession.** This Commercial Item Description governs procurement of IC/SM switchboards, which were previously a part of MIL-A-17196(SH). Procurement of all other components described in MIL-A-17196E(SH) shall be in accordance with that document or its applicable successor.

2. **Identification numbers.** The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



3. **Ordering data.** Procurement documents should specify the following:

a. IC/SM Alarm Switchboard configuration.
b. Manufacturers name and catalog identification number if required.
c. Finish color. (Composite switchboards will require tinting)
d. Required delivery date.

CIVIL AGENCY COORDINATING ACTIVITIES:
COMMERCE - NIST
GSA-FSS

Preparing Activity:
Navy - SH
Project (6320-N038)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER CID-A-A-50623	2. DOCUMENT DATE (YYMMDD) 16 September 1994

3. DOCUMENT TITLE
ALARM SWITHCHBOARD TYPE IC/SM

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical point of contact: Mr. Walston, 03K32	b. TELEPHONE (Include Area Code) (1) Commercial 703 602-7241 (2) AUTOVON 332-7241
ADDRESS (Include Zip Code) Commander SEA 03R42 2531 Jefferson Davis Hyway Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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